

INTRODUCTION TO OBJECT ORIENTED PROGRAMMING (OOP)

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Lecture 1

OOP

Why We Need OOP

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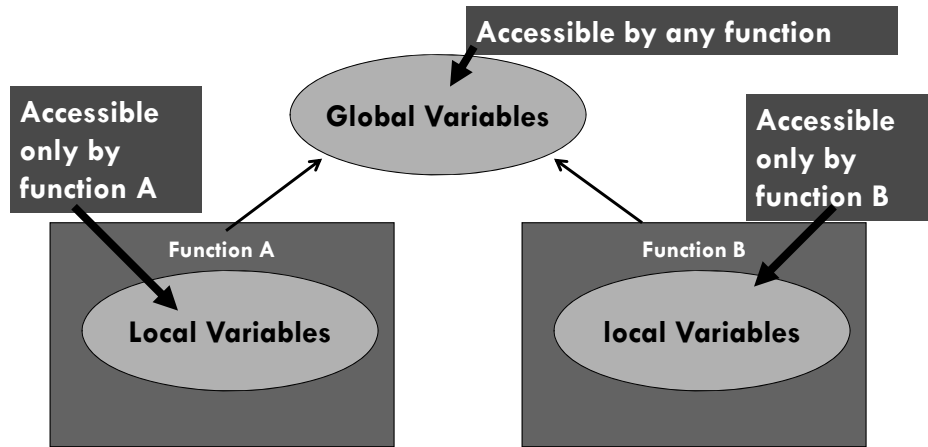
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- OOP was developed to overcome the limitations of the traditional programming languages.
- As the programs gets larger in procedural programming it becomes more complex to understand the logic and implement.

Problems with Structured Approach

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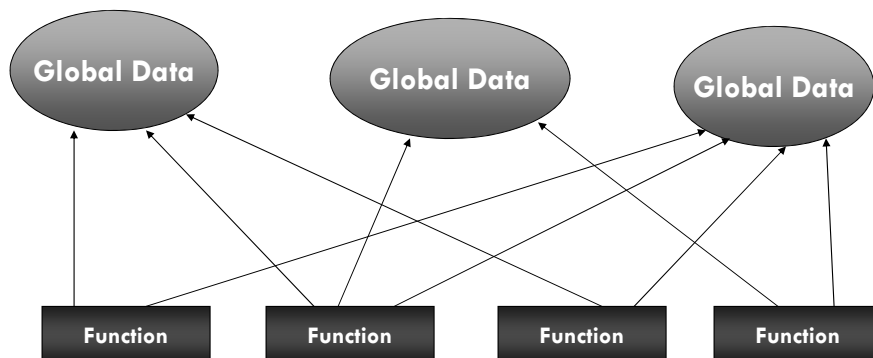
- Unrestricted access



Global and local variables.

Problems with structured approach

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The procedural paradigm

Object Oriented Approach

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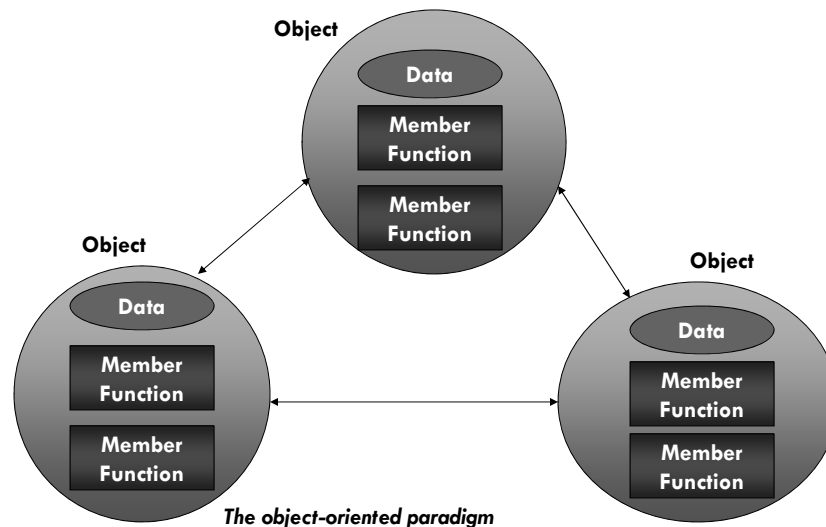
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- The fundamental idea behind **object-oriented language** is to combine into a single unit both data and the functions that operate on that data. Such a unit is called an object
- Data and its functions are said to be encapsulated into a single entity
- Data encapsulation and data hiding are the key terms in the description of object oriented languages

Object Oriented Approach

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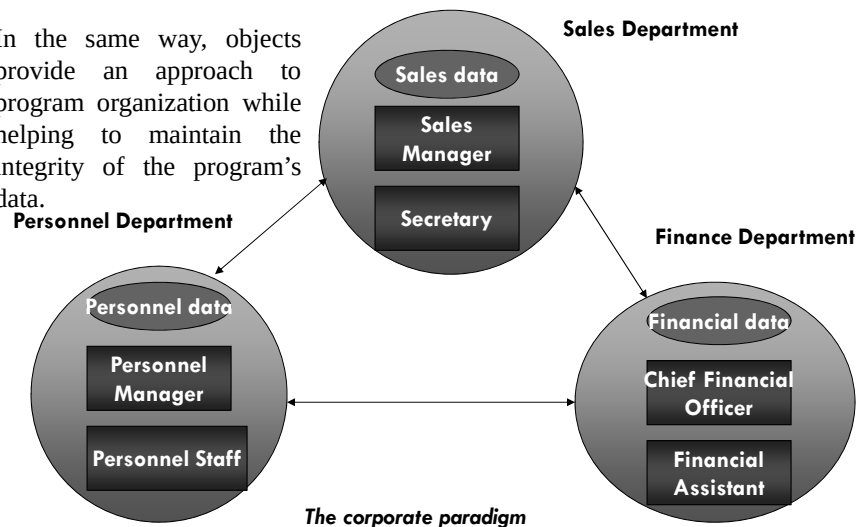
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OOP: An Approach to Organization

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- In the same way, objects provide an approach to program organization while helping to maintain the integrity of the program's data.



What kinds of things become objects in object-oriented programs?

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• Physical objects

Automobiles in a traffic-flow simulation; Electrical components in a circuit-design program; Countries in an economics model; Aircraft in an air traffic control system

• Components in computer games

Cars in an auto race; Positions in a board game (chess, checkers); Animals in an ecological simulation; Opponents and friends in adventure games

• Collections of data

An inventory, A personnel file, A dictionary, A table of the latitudes and longitudes of world cities

• Data-storage constructs

Customized arrays, Stacks, Linked lists, Binary trees

• Human entities

Employees, Students, Customers, Salespeople

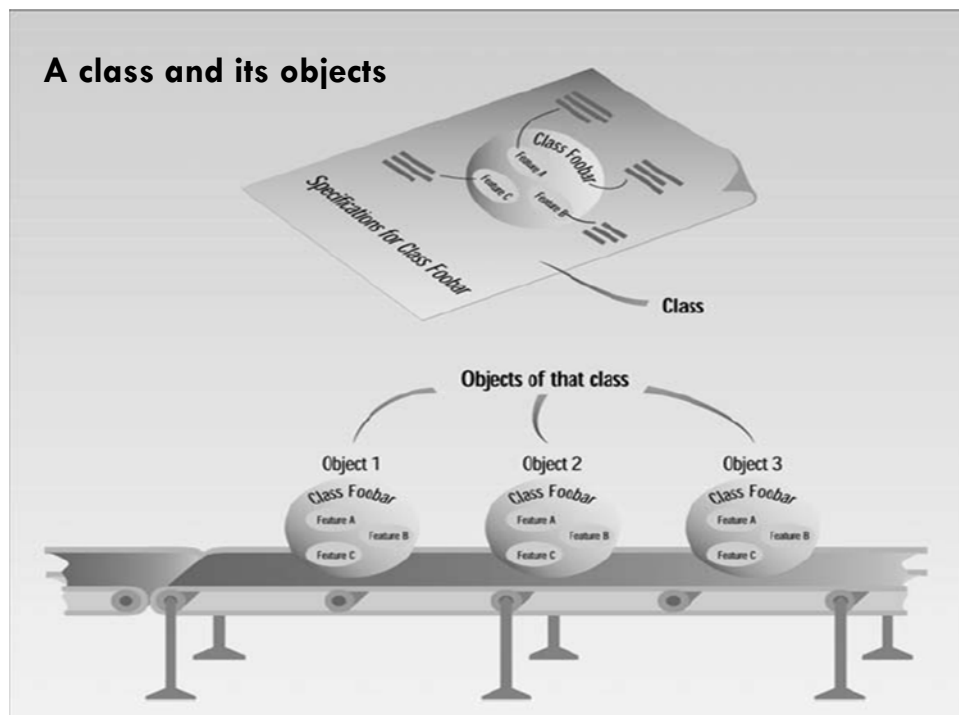
Classes

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In OOP we say that objects are members of classes.

Classes provide modularity and structure in an object-oriented computer program. A class should typically be recognizable to a non-programmer familiar with the problem domain, meaning that the characteristics of the class should make sense in context. Also, the code for a class should be relatively self-contained (generally using encapsulation). Collectively, the properties and methods defined by a class are called members.



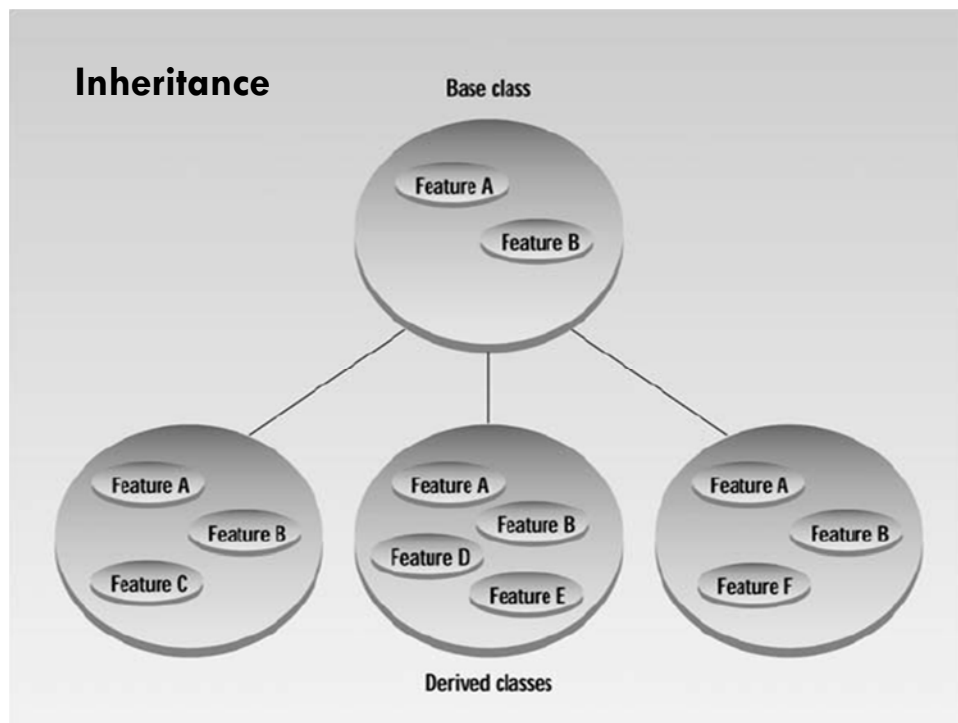
Inheritance

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The idea of classes leads to the idea of inheritance. In our daily lives, we use the concept of classes divided into subclasses. For examples:

- ❑ The animal class is divided into mammals, amphibians, insects, birds, and so on.
- ❑ The vehicle class is divided into cars, trucks, buses, motorcycles, and so on.
- ❑ In C++ the original class is called the **base class**; other classes can be defined that share its characteristics, but add their own as well. These are called **derived classes**.
- ❑ **Derived classes** inherit some characteristics from their base class, but add new ones of their own.



A Simple Class

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- The program demonstrates the syntax and general features of classes in C++. The program contains a class and two objects of that class. Here's the listing for the SMALLOBJ program:

```
// smallobj.cpp
#include <iostream>
using namespace std;
////////////////////////////////////
class smallobj //define a class
{
    private:
        int somedata; //class data
    public:
        void setdata(int d) //member function to set data
        { somedata = d; }
        void showdata() //member function to display data
        { cout << "Data is " << somedata << endl; }
};
////////////////////////////////////
int main()
{
    smallobj s1, s2; //define two objects of class smallobj
    s1.setdata(1066); //call member function to set data
    s2.setdata(1776);
    s1.showdata(); //call member function to display data
    s2.showdata();
    return 0;
}
```

References

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- Object Oriented Programming in C++ by Robert Lafore, 2002 by Sams Publishing.
- C++ How to Program 5th edition by Deitel and Deitel, Prentice Hall 2005